

# Simplify Algebraic Expressions

Lesson 6-15

**Name:**

**Date:**

**Class:**

---



## TODAY'S OBJECTIVES

**Content Objective:** I can simplify algebraic expressions by combining like terms.

**Language Objective:** I can explain my steps using the words like terms, combine, simplify, and coefficient.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



**Fill in each blank as we go. Use the Word Bank to help you.**



## WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Simplify Algebraic Expressions

Like Terms

Combine

Simplify

Coefficient

Term

Distributive property



Tap any word to see what it means and a picture.

Level 3: try the blanks from memory first, then check the bank.

1

— Rewrite an expression in a shorter equivalent form by combining like terms and using properties.

2

Terms with the same variable part, like  $3y$  and  $8y$ , are

.

3

To add or subtract like terms into a single term is to

them.

4

To write an expression in its shortest equal form is to

it.

5

The number multiplied by a variable is the .

6

A number or variable separated by + or – signs is a .

7

Multiplying each part of a sum by a number uses the  property.

## Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

### 1. Understand the Question

EXPLAIN

**Simplify the total order expression. What does each part of the simplified expression tell you?**

**Your job:** Answer the question. Use math evidence. Explain how the evidence proves your answer.

*Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.*

## 2. Plan Your Math Words

Check at least two words you will use.

- ☐ **Simplify Algebraic Expressions** — Rewrite an expression in a shorter equivalent form by combining like terms and using properties.  
*Simplificar expresiones algebraicas — Reescribir una expresión en una forma equivalente más corta combinando términos semejantes y usando propiedades.*

- ☐ **Like Terms** — Terms with the same letter raised to the same power, like  $2x$  and  $5x$ .  
*Términos semejantes — Términos con la misma letra elevada a la misma potencia, como  $2x$  y  $5x$ .*

- ☐ **Combine** — To add or subtract terms with the same letter into one.  
*Combinar — Sumar o restar términos con la misma letra en uno solo.*

- ☐ **Simplify** — To write an expression in its shortest form.  
*Simplificar — Escribir una expresión en su forma más corta.*

- ☐ **Coefficient** — The number in front of a letter, like the 3 in  $3x$ .  
*Coeficiente — El número frente a una letra, como el 3 en  $3x$ .*

**Say it first:** Say your idea to a partner or quietly to yourself before you write.

*Di tu idea a un compañero o en voz baja antes de escribir.*

**The total order simplifies to \_\_\_\_m + \_\_\_\_.** The \_\_\_\_ tells us \_\_\_\_ and the \_\_\_\_ tells us \_\_\_\_.

*Di tu respuesta primero: Mi respuesta es \_\_\_\_ porque \_\_\_\_.*

### 3. Build Your Explanation

**Model the parts:** This model shows the parts of an explanation. It does not answer your writing question.

**Claim:** You can only combine items of the same type, just like like terms.

**Evidence:** Students explain microphones combine with microphones ( $4 + 2 + 6 = 12$ ) but not with stands, connecting this to combining like terms in algebra.

**Reasoning:** This evidence connects the definition of simplify algebraic expressions to the mathematical claim.

#### Start

Most language support

Complete the frames. Use the word bank.

*Completa las oraciones. Usa el banco de palabras.*

- **The total order simplifies to** \_\_\_\_ **m +** \_\_\_\_ **. The** \_\_\_\_ **tells us** \_\_\_\_ **and the** \_\_\_\_ **tells us** \_\_\_\_.
  - **My answer is** \_\_\_\_.
- Mi respuesta es* \_\_\_\_.

---

---

---

---

#### Build

Some language support

Write two connected sentences. Use because, so, or but.

*Escribe dos oraciones conectadas. Usa porque, entonces o pero.*

- **My claim is** \_\_\_\_.
- Mi afirmación es* \_\_\_\_.
- **I know because** \_\_\_\_.
- Lo sé porque* \_\_\_\_.
- **So** \_\_\_\_.
- Entonces* \_\_\_\_.

---

---

---

## Explain

### Light language support

Write a claim, give math evidence, and explain your reasoning.

*Escribe una afirmación, da evidencia matemática y explica tu razonamiento.*

- **My claim is \_\_\_\_.**  
*Mi afirmación es \_\_\_\_.*
- **My math evidence is \_\_\_\_.**  
*Mi evidencia matemática es \_\_\_\_.*
- **This proves \_\_\_\_ because \_\_\_\_.**  
*Esto demuestra \_\_\_\_ porque \_\_\_\_.*

---

---

---

---

---

## 4. Check Your Explanation

- ☐ I answered the question.  
*Respondí la pregunta.*
- ☐ I used math evidence: numbers, an equation, a model, or a comparison.  
*Usé evidencia matemática: números, una ecuación, un modelo o una comparación.*
- ☐ I explained how the evidence proves my answer.  
*Explicué cómo la evidencia demuestra mi respuesta.*
- ☐ I used at least two math words.  
*Usé por lo menos dos palabras matemáticas.*
- ☐ I reread complete sentences and punctuation.  
*Volví a leer mis oraciones completas y la puntuación.*